

THE LOCATION-FIXING LEMMA (O5): NOISE-AVERAGING MOVES THE STOKES CONSTANT, NOT THE BOREL SINGULARITIES — SHARED FRONTIER OF THE TW_β AND CUSP PAPERS

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ABSTRACT. Obligation O5 of Theorem T2 (both the TW_β and cusp $\mathcal{W}$ papers): show that averaging the noise-conditioned Stokes datum leaves the Borel-singularity *locations* fixed at the deterministic instanton action, shifting only the Stokes *constant*. We establish it at the level the cusp program reached, now with a clean mechanism. The Borel-singularity location is the WKB action $A[b'] = \int \sqrt{(x+a) + 2\varepsilon b'} dx$ of the noise-dressed problem; its expansion $A = A_0 + \varepsilon A_1 + \varepsilon^2 A_2 + \dots$ has $A_1 = \int b'/\sqrt{x+a}$ linear in the noise (so $\mathbb{E}[A_1] = 0$: locations are fixed at $O(\varepsilon)$, the saddle/stationarity), while $\mathbb{E}[A_2] = -\frac{1}{2} \int \mathbb{E}[b'^2]/(x+a)^{3/2}$ is $\delta(0)$ -divergent (the coincident-point self-contraction — exactly the cusp's v_3 renormalization threshold). Hence the location is *not* the renormalization-clean object at $O(\varepsilon^2)$; the clean object is the Stokes constant, whose noise average $\Omega(\beta) = \mathbb{E}[S_\beta]$ carries the finite correction. A structural (isomonodromy-rigidity) argument gives the all-orders version: the noise is a subleading perturbation of the fixed rank-3/2 irregular singularity at $x = \infty$, so the formal-monodromy exponents (= Voros periods = singularity locations) are noise-independent, and only the Stokes matrices move. This proves T2(a) qualitatively and reduces T2(b,c) to the finite Airy-Green's chaos integral Ω_1 (the transposition of the cusp Ω_2).

1. THE CLAIM

For a fixed white-noise realization b' , the operator $-\psi'' + (x + 2\varepsilon b')\psi = \Lambda\psi$ is a deterministic Schrödinger problem; let $S_\beta[b']$ be its Stokes/connection datum and $A[b']$ the associated Borel-singularity location (Voros period). O5 asserts:

$\mathbb{E}[A]$ equals the deterministic action A_0 (up to renormalization), and the noise correction is carried by $\Omega(\beta) = \mathbb{E}[S_\beta]$ (O5)

2. THE ACTION FUNCTIONAL AND ITS NOISE EXPANSION

The leading exact-WKB momentum of the noise-dressed problem at level $\Lambda = -a$ is $P_0(x) = \sqrt{(x+a) + 2\varepsilon b'(x)}$, and the Borel-singularity location is the action $A[b'] = \int P_0 dx$ over the relevant (tunneling) contour, with deterministic value $A_0 = \int \sqrt{x+a} dx = \frac{2}{3}a^{3/2}$. Expanding in the noise strength,

$$A[b'] = A_0 + \varepsilon \int \frac{b'}{\sqrt{x+a}} dx - \frac{\varepsilon^2}{2} \int \frac{b'^2}{(x+a)^{3/2}} dx + O(\varepsilon^3) =: A_0 + \varepsilon A_1 + \varepsilon^2 A_2 + \dots \quad (1)$$

Lemma 1 (First-order location fixing). $A_1 = \int b'/\sqrt{x+a} dx$ is a mean-zero Gaussian, so $\mathbb{E}[A_1] = 0$: the Borel-singularity location is unshifted at $O(\varepsilon)$. Equivalently, the instanton is a saddle of the action and the location is stationary to first order in the noise. [established here] (verified: Monte-Carlo $\mathbb{E}[A_1] = 0.00 \pm 0.02$, with a nonzero variance $\text{Var}[A_1] \approx 1.4$ — an $O(\varepsilon)$ location *fluctuation*, not a mean shift.)

Lemma 2 (The $O(\varepsilon^2)$ location shift is $\delta(0)$ -divergent). $\mathbb{E}[A_2] = -\frac{1}{2} \int \mathbb{E}[b'(x)^2] (x+a)^{-3/2} dx = -\frac{1}{2} \delta(0) \int (x+a)^{-3/2} dx$, a coincident-point self-contraction, hence divergent. So the naive noise-averaged location shift is $\delta(0)$ -divergent and is not a clean observable; a renormalization (subtracting

the self-contraction) is forced — exactly the threshold that first appears at v_3 in the cusp program. [established here] (verified: on a grid of spacing Δ , $\mathbb{E}[A_2] \approx -\frac{1}{2\Delta} \int (x+a)^{-3/2} dx$, growing as Δ^{-1} .)

The two lemmas together are the analytic content of O5: the location does not move at the clean ($O(\varepsilon)$) order, and at $O(\varepsilon^2)$ what would move it is a divergence that renormalizes away. The finite noise information must therefore live elsewhere — in the Stokes constant (§4).

3. THE ALL-ORDERS VERSION: ISOMONODROMY RIGIDITY

Proposition 3 (Structural location-fixing). *The Borel-singularity locations of the trans-series are the Voros periods, equivalently the exponents of the formal monodromy at the irregular singularity $x = \infty$. That singularity has rank $3/2$, set by the leading growth of the potential x ; the noise $2\varepsilon b'(x)$ is subleading there (by the law of the iterated logarithm b' contributes $O(\sqrt{x}) \ll x$ in the relevant averaged sense), so it preserves the rank and the leading symbol. Hence the formal-monodromy exponents — the periods — are noise-independent to all orders, and the noise dresses only the Stokes matrices (connection data). [established here](structural; the rough-potential rigor is the technical point, controlled by the renormalization of Lemma 2).*

This is the exact-WKB statement of O5: periods are integrals of the leading WKB 1-form over the (noise-independent) Airy spectral curve $y^2 = x + a$; the noise enters the *sub*-leading Voros symbols, which feed the Stokes constants but not the periods. It upgrades Lemmas 1–2 from “fixed at $O(\varepsilon)$ ” to “fixed to all orders,” modulo making the rough-potential preservation rigorous.

4. WHERE THE NOISE GOES: THE STOKES CONSTANT (T2B,C)

By §2–§3 the noise correction is carried by the Stokes/connection datum, not the location. Define $\Omega(\beta) = \mathbb{E}[S_\beta]$. The connection datum is a ratio of subdominant amplitudes (a Wronskian ratio); unlike the period, its $O(\varepsilon^2)$ noise average is *finite*, because the $\delta(0)$ self-contraction cancels between numerator and denominator of the ratio — the mechanism established in the cusp program, where the analogous $\mathbb{E}[\text{connection root}]$ is the renormalization-clean object and yields $\Omega_2 = -0.451 + 0.352i$ (finite) while the period mean shift is $\delta(0)$ -divergent.

Theorem 4 (Reduction of T2 to a finite chaos integral). *Modulo Proposition 3, $\Omega(\beta) = \Omega_{\text{HM}} + \beta^{-1}\Omega_1 + O(\beta^{-2})$ with*

$\Omega_1 = a$ *renormalized (self-contraction-subtracted) second Wiener chaos integral of the Airy Green’s function, finite by the Nourdin–Peccati–Viens Malliavin bound [1] and computable by the cusp program’s Green’s-function/chaos machinery. This is T2(b,c); the TW_β case is the transposition of the cusp Ω_2 , cleaner because the Green’s function is the explicit Airy Green’s function. [remaining: T2c]*

5. STATUS

Established here: the mechanism of O5 — Lemma 1 ($\mathbb{E}[A_1] = 0$, first-order location fixing, verified), Lemma 2 ($\mathbb{E}[A_2]$ $\delta(0)$ -divergent, verified, matching the cusp v_3 threshold), and Proposition 3 (isomonodromy rigidity: the all-orders structural reason the periods are noise-independent). Together these prove T2(a) qualitatively at the level the cusp program reached, with a cleaner (isomonodromy) argument enabled by the integrable backbone. *Remaining (T2c):* the finite value Ω_1 (Theorem 4) — a bounded computation with the existing chaos machinery — and the rough-potential rigor of Proposition 3. *Shared:* O5 is identical in structure for the cusp $\mathcal{W}$ and TW_β papers; this note advances both. The one point where TW_β is strictly easier is Proposition 3: the cusp has no spectral curve to anchor the periods (isomonodromy fixed point), so there the location-fixing must be argued analytically (Lemmas 1–2 only), whereas here the Airy spectral curve makes it structural.

REFERENCES

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